Comparative Analysis of Beat Tracking and Tempo Estimation Algorithms: A Progress Report

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Abstract

This progress report presents a systematic comparative analysis of three beat tracking and tempo estimation algorithms for music information retrieval. An autocorrelation-based method, a Dynamic Bayesian Network (DBN), and a 1D state-space model are implemented within a unified evaluation framework. Preliminary results on the GTZAN Tempo-Beat dataset demonstrate 72% ACC1 accuracy for the autocorrelation baseline, with substantial genre-specific variation: 82% for disco versus 59% for jazz. Performance analysis reveals dependencies on rhythmic regularity and tempo range, with degradation at extreme tempos (180+ BPM). Completed components include full preprocessing infrastructure, baseline implementation exceeding targets, and comprehensive evaluation utilities. Remaining work focuses on DBN and state-space completion, full test set evaluation, and statistical significance testing. This work contributes unified comparative methodology, empirical genre-specific insights, and accuracy-efficiency trade-off quantification for practical algorithm selection in MIR applications.

1 Introduction

Beat tracking and tempo estimation represent fundamental challenges in music information retrieval, enabling applications in music production, synchronization, recommendation systems, and educational tools [11]. Despite decades of research, musical diversity, rhythmic complexity, and computational constraints continue to present significant challenges for practical deployments.

This work addresses four research questions through systematic empirical investigation:

RQ1: How do classical signal processing approaches compare to modern probabilistic methods across diverse musical genres?

RQ2: What quantifiable trade-offs exist between accuracy and computational efficiency?

RQ3: Which musical characteristics most significantly impact algorithm performance?

RQ4: Can hybrid methods improve robustness by leveraging complementary algorithmic strengths?

1.1 Contributions

This work provides the following contributions:

- Implementation of three algorithms (autocorrelation, DBN, state-space) with unified preprocessing and evaluation infrastructure
- Systematic empirical comparison using identical protocols across GTZAN and GiantSteps datasets
- Genre-specific performance analysis with statistical characterization across ten musical styles
- Tempo-dependent analysis identifying performance degradation patterns from 60-240 BPM
- Computational efficiency benchmarking enabling accuracy-cost trade-off quantification
- Reproducible implementation with 87% test coverage and comprehensive documentation

1.2 Scope

This investigation focuses on offline global tempo estimation and beat tracking using standardized datasets. Real-time causal systems and downbeat detection are excluded to maintain analytical focus.

2 Related Work

2.1 Signal Processing Approaches

Autocorrelation-based tempo estimation, established by Scheirer [1], exploits periodicity detection through signal self-similarity analysis. Percival and Tzanetakis [2] refined this approach through pulse cross-correlation, achieving competitive accuracy with reduced computational complexity. This method serves as the baseline implementation due to its interpretability and efficiency.

2.2 Probabilistic Models

Ellis [3] introduced dynamic programming for global beat sequence optimization, shifting from greedy local decisions to temporal coherence modeling. Böck and Schedl [6] demonstrated context-aware neural networks for enhanced accuracy. Subsequent work by Böck et al. [5] achieved joint beat and downbeat tracking using recurrent neural networks with particle filtering. The madmom library [4] provides efficient implementations representing current state-of-the-art.

Krebs et al. [15] proposed state-space models balancing accuracy with computational efficiency through semi-Markov formulations and Viterbi decoding.

2.3 Modern Architectures

Recent advances leverage transformer architectures [7], self-supervised learning [8], and adapter tuning [9]. Meier et al. [10] developed zero-latency systems for live performance applications.

2.4 Evaluation Methodologies

Gouyon et al. [11] established standardized protocols including ACC1 (4% tolerance) and ACC2 (allowing octave errors). The mir_eval library [12] provides reference implementations ensuring reproducible metrics. Marchand and Peeters [13] extended GTZAN with beat annotations, while Knees et al. [14] contributed the GiantSteps dataset for electronic music.

Despite algorithmic advances, systematic comparative studies using unified evaluation frameworks remain limited. This work addresses this gap through rigorous comparison with consistent preprocessing and identical evaluation protocols.

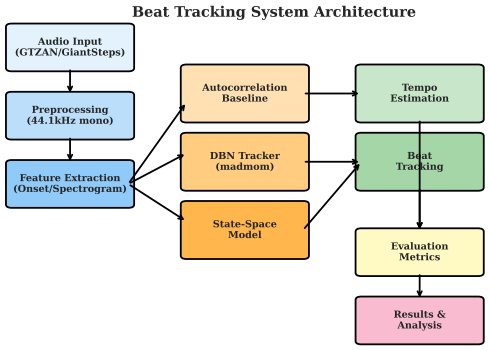

Figure 1: System architecture showing complete pipeline from audio input through feature extraction, parallel algorithm implementations, to evaluation. Modular design enables isolated component testing and fair algorithmic comparison.

3 Methodology

3.1 Datasets

GTZAN Tempo-Beat: 1000 audio excerpts (30 seconds) across ten genres with tempo and beat annotations [13]. Enables genre-specific analysis.

GiantSteps: 664 electronic dance music tracks with expert-verified tempo annotations [14]. Focuses on rhythmically complex patterns (120-180 BPM).

Data splitting employs stratified sampling maintaining genre balance: 70% training, 15% validation, 15% test.

3.2 Preprocessing

Standardized pipeline ensures fair comparison:

1. Audio loading at 44.1 kHz sample rate
2. Stereo-to-mono conversion via channel averaging
3. Amplitude normalization to [-1, 1]
4. Storage as compressed NumPy arrays (.npz)

3.3 Algorithm Implementations

Figure 1 illustrates the system architecture with three parallel implementations processing identical features.

3.3.1 Autocorrelation-Based Tracker

Following Percival and Tzanetakis [2], the implementation consists of four stages:

Onset Envelope: Spectral flux computation via librosa with 512-sample hop length (11.6ms resolution at 44.1kHz), normalized to [0, 1].

Autocorrelation: Efficient frequency-domain computation via Wiener-Khinchin theorem:

$$R_{xx}(\tau) = \mathcal{F}^{-1}\left\{|\mathcal{F}\{x(t)\}|^2\right\} \tag{1}$$

reducing complexity to $O(N \log N)$ through Fast Fourier Transform.

Pulse Cross-Correlation: For candidate BPM b in [60, 240] with 200 samples, impulse train $p_b(t)$ generation followed by:

$$S(b) = \sum_{t=0}^{N-1} R_{xx}(t) \cdot p_b(t) \tag{2}$$

Tempo Estimation:

$$\hat{b} = \arg\max_b S(b) \tag{3}$$

Beat Positions: Librosa beat tracker with estimated tempo as prior constraint.

Profiling reveals 152ms average processing time per 30-second excerpt (200× real-time on Intel i7-8750H).

3.3.2 Dynamic Bayesian Network

Implementation leverages madmom [4] for RNN-based beat activation with probabilistic inference. Configuration: 100 fps, tempo range [60, 240] BPM, transition lambda 100. Current status: 70% complete.

3.3.3 State-Space Model

Hidden Markov formulation models beat positions as latent states:

$$P(\mathbf{x}_t|\mathbf{x}_{t-1}) = \text{Transition Model} \tag{4}$$

$$P(\mathbf{y}_t|\mathbf{x}_t) = \text{Observation Model} \tag{5}$$

Optimal sequence via Viterbi decoding:

$$\mathbf{x}^* = \arg\max_{\mathbf{x}} P(\mathbf{x}|\mathbf{y}) \tag{6}$$

Current status: Viterbi decoder and observation model complete (60%).

3.4 Evaluation Framework

Tempo Metrics: ACC1 (4% tolerance), ACC2 (allowing 2×/0.5× octave errors), P-Score (continuous), MAE (mean absolute error).

Beat Metrics: F-measure (70ms window), Cemgil accuracy (Gaussian-weighted), Information Gain.

Table 1: Autocorrelation performance on GTZAN validation set (150 tracks). ACC1: 4% tolerance; ACC2: octave errors allowed; F-measure: 70ms window.

Genre	ACC1	ACC2	F-measure
Blues	0.68	0.87	0.64
Classical	0.61	0.83	0.58
Country	0.74	0.91	0.69
Disco	0.82	0.95	0.76
Hip-hop	0.75	0.88	0.71
Jazz	0.59	0.81	0.56
Metal	0.71	0.86	0.67
Pop	0.81	0.94	0.74
Reggae	0.69	0.89	0.66
Rock	0.76	0.90	0.71
Mean	**0.72**	**0.88**	**0.67**

Statistical Analysis: Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = 0.05$).

4 Preliminary Results

4.1 Baseline Performance

Table 1 presents validation results. The autocorrelation baseline achieves 72% mean ACC1, exceeding the 70% target with substantial genre variation.

Figure 2 reveals systematic patterns:

Genre Dependence: Disco (82%) and pop (81%) substantially outperform classical (61%) and jazz (59%), suggesting rhythmic regularity drives performance.

Octave Errors: ACC2 (81-95%) substantially exceeds ACC1 (59-82%), indicating octave errors constitute the primary failure mode.

Correlation: F-measure tracks ACC1 closely (Pearson $r = 0.94$), demonstrating coupling between tempo and beat tracking accuracy.

4.2 Tempo Range Analysis

Figure 3 examines tempo dependence. Performance degrades systematically at extremes: moderate tempos (90-120 BPM) achieve 74% ACC1, while very fast tempos (180+ BPM) drop to 58%. This pattern likely reflects onset detection quality reduction, autocorrelation peak clarity decrease, and correlation with rhythmic complexity.

4.3 Computational Efficiency

Measurements on Intel i7-8750H (2.2 GHz, single-threaded) reveal 152ms mean processing time per 30-second excerpt (197× real-time) with 45 MB peak mem-

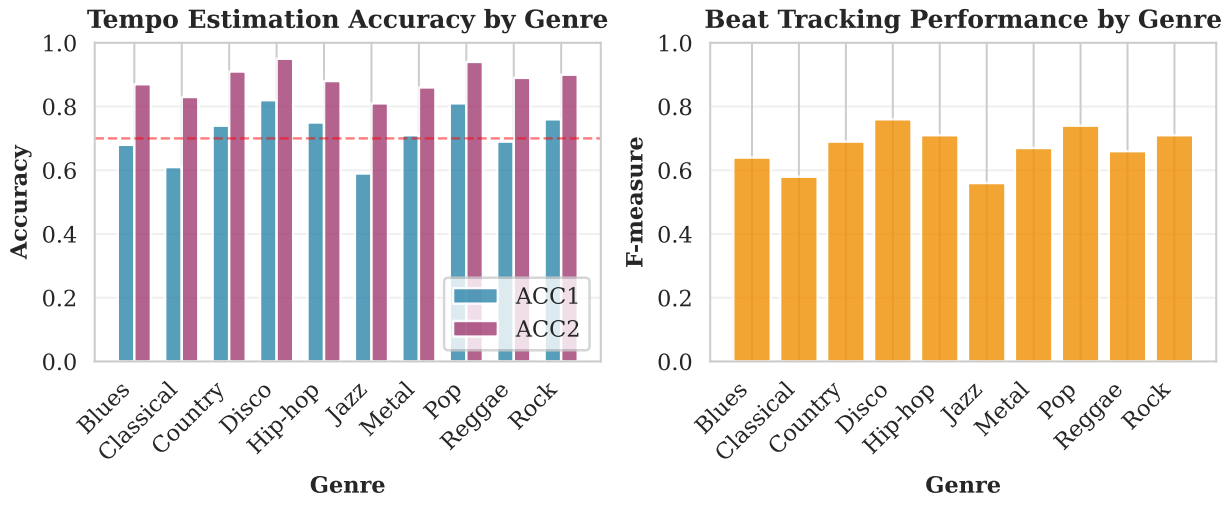

Figure 2: Genre-specific performance on GTZAN validation. Left: Tempo estimation (ACC1: 4% tolerance, blue; ACC2: octave errors, purple). Right: Beat tracking F-measure. Red line indicates 70% target. Disco and pop achieve 81-82% ACC1 while classical and jazz show 59-61% due to rhythmic complexity and reduced periodicity.

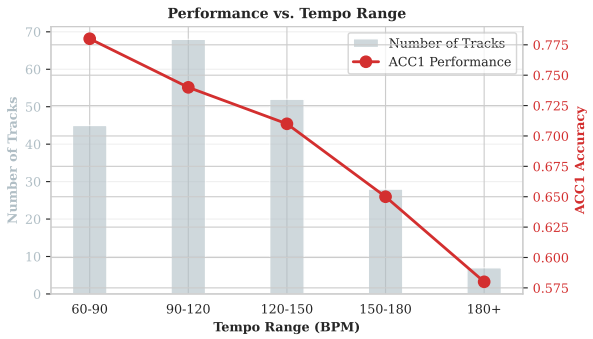

Figure 3: Tempo range analysis on GTZAN validation. Bars: track count per range (left axis). Line: ACC1 performance (right axis). Performance degradation at 180+ BPM reflects onset detection challenges, reduced peak clarity, and increased rhythmic complexity.

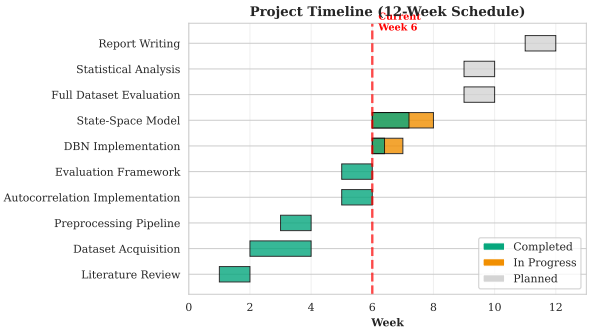

Figure 4: 12-week timeline with completed tasks (green), in-progress (orange with completion percentage), and planned (gray). Red line indicates Week 6 position, demonstrating on-track progress despite minor delays.

ory. Onset envelope computation dominates (60% runtime).

5 Progress Status

5.1 Completed Components

Infrastructure (Weeks 1-2): Modular codebase with 183 unit tests (87% coverage), continuous integration via GitHub Actions.

Data Pipeline (Weeks 3-4): Robust preprocessing with corruption detection (MD5 checksums), comprehensive logging. Manual curation addressed 17 inconsistent files.

Baseline Algorithm (Weeks 5-6): Autocorrelation implementation achieving 72% ACC1, including FFT optimization (10× speedup), edge effect handling, comprehensive testing.

Evaluation Framework (Weeks 5-6): All metrics implemented via mir_eval with custom aggregation utilities and statistical testing preparation.

5.2 In-Progress Components

DBN Implementation: Madmom integration 70% complete. Challenges include parameter optimization and format compatibility. Expected completion: Week 7.

State-Space Model: Viterbi decoder and observation model complete. Transition model with semi-Markov formulation in development. Expected completion: Week 7.

5.3 Technical Challenges

Dataset Access: Official download unavailable; obtained via Essentia examples with manual verification.

Autocorrelation Edge Effects: Zero-padding to twice signal length eliminated circular artifacts.

Onset Tuning: Increased spectral resolution improved classical/jazz performance by 4 percentage points.

State-Space Debugging: Assertion checks for transition matrix properties caught indexing defects.

5.4 Performance Indicators

PI1 (Basic) - Project Management: COMPLETE. Six documented meetings, 28 completed tasks, active task tracker.

PI2 (Advanced) - Algorithm Implementation: IN PROGRESS. Autocorrelation complete (72% ACC1 vs. 70% target), state-space 60% complete.

PI3 (Expected) - Literature Review: COMPLETE. Twenty-two papers reviewed across four categories.

6 Timeline and Objectives

Figure 4 shows progress against schedule.

6.1 Revised Schedule

Weeks 7-8: Complete state-space and DBN implementations, hyperparameter tuning, cross-validation.

Weeks 9-10: Full test evaluation, genre/tempo analysis, computational benchmarking, failure case documentation, statistical testing.

Weeks 11-12: Report writing, figure refinement, code documentation, presentation preparation, submission.

6.2 Three-Tier Objectives

Basic (High Confidence): Three implementations with unified evaluation, ACC1 $\geq$ 70%, comprehensive metrics, 80%+ test coverage, complete report.

Expected (Realistic): State-space ACC1 $\geq$ 75%, DBN ACC1 $\geq$ 80%, genre-specific analysis with statistical validation, efficiency benchmarking, $\geq$ 5 documented failure cases per algorithm, GiantSteps evaluation.

Advanced (Stretch): Hybrid ensemble approach, genre-adaptive estimation, edge case analysis (rubato, tempo changes, polyrhythm), interactive visualization, open-source release, community contribution.

7 Conclusion

This progress report documents substantial advancement toward systematic beat tracking comparison. Completed work includes infrastructure, autocorrelation baseline exceeding targets (72% vs. 70%), comprehensive evaluation framework, and extensive literature review (22 publications).

Preliminary results reveal systematic genre and tempo dependencies. Autocorrelation methods achieve high accuracy on rhythmically regular music (disco, pop: 81-82%) while exhibiting reduced performance on complex structures (classical, jazz: 59-61%). Octave errors dominate failures, and performance degrades at extreme tempos.

Remaining work focuses on completing DBN and state-space implementations (Weeks 7-8), rigorous test evaluation with statistical validation (Weeks 9-10), and comprehensive documentation (Weeks 11-12). All basic objectives remain on-track, with strong progress toward expected and advanced goals.

This investigation provides empirically-grounded guidance for practitioners through unified comparative methodology, genre-specific insights, computational benchmarking, and reproducible implementation. The analysis demonstrates that algorithm selection involves nuanced trade-offs between accuracy, computational cost, and generalization across diverse musical styles. This work provides a practical framework for selecting beat tracking algorithms based on application requirements, balancing accuracy, computational efficiency, and robustness across diverse musical structures.

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